

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 2.6: SURFACE AREA OF SOLIDS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Final Explanation: Is the shopkeeper right? A hardware shop along Kirinyaga Road in Nairobi sells three 10-litre metal buckets: a cylindrical bucket, a frustum-shaped (tapered) bucket and a conical bucket. The shop owner says the tapered bucket is the most expensive because it uses the most metal. Use what you learned in Lessons 1 to 8 to answer the driving question: How do we calculate exactly how much material is needed to cover or construct any solid shape, and why does the shape of a container matter even when the volume stays the same? Answer every section in the space below it. Show your working, write units (cm, cm², m²) and use $\pi = 3.142$ unless told otherwise. Use diagrams or nets where they help you explain. Give reasons in full sentences, not just answers.

Section 1: The cone and its net

A conical paper cup like those at the water dispensers in Nairobi hospitals has base radius $r = 3.5$ cm and slant height $l = 8$ cm.

(a) Describe what shape the curved part of the cone becomes when you unfold it, and explain how the arc length of this shape is related to the base of the cone.

(b) Explain why the lateral (curved) area of a cone is πrl . Use the idea that the sector is a fraction of a full circle of radius l .

(c) Calculate the total surface area of the cup.

(d) Explain why you must use the slant height l and not the perpendicular height h .

(a) The curved part unfolds into a sector of a circle whose radius is the slant height l . The arc length of the sector is equal to the circumference of the circular base of the cone, so $\text{arc} = 2\pi r$.

(b) A full circle of radius l has circumference $2\pi l$ and area πl^2 . The sector has arc $2\pi r$, so it is the fraction $2\pi r / 2\pi l = r/l$ of the full circle. Its area is $(r/l) \times \pi l^2 = \pi rl$.

(c) $\text{TSA} = \pi rl + \pi r^2 = \pi(3.5)(8) + \pi(3.5)^2 = 28\pi + 12.25\pi = 40.25\pi \approx 40.25 \times 3.142 \approx 126.5 \text{ cm}^2$.

(d) The curved surface is the sector, and the sector's radius is the slant height l , measured along the surface. The perpendicular height h goes through the inside of the cone and is not a length on the metal or paper surface, so using it would give the wrong area.

Section 2: The tapered (frustum) bucket

The tapered bucket on Kirinyaga Road is open at the top. Its top radius is $R = 14$ cm, its base radius is $r = 8$ cm and its perpendicular height is $h = 25.7$ cm.

(a) Describe the net of the curved surface of this bucket and explain why it looks larger and more complex than the net of a cylinder.

(b) Calculate the slant height l .

(c) Calculate the amount of sheet metal needed for the open bucket (curved surface plus base only).

(d) Explain, using the idea of a large cone with a small cone removed, why the curved surface area of a frustum is $\pi(R + r)l$.

(e) How would your working change if the bucket had a lid?

(a) The curved surface of a frustum unfolds into an annular sector, which is the region between two concentric arcs. It is a curved ring-shaped piece, not a simple rectangle. Its area depends on both radii (R and r) and on the slant height at the same time, so the net looks larger and more complex than the rectangle that forms the curved surface of a cylinder.

(b) $l = \sqrt{h^2 + (R - r)^2} = \sqrt{25.7^2 + (14 - 8)^2} = \sqrt{660.49 + 36} = \sqrt{696.49} \approx 26.4$ cm.

(c) Open bucket $SA = \pi(R + r)l + \pi r^2 = \pi(14 + 8)(26.4) + \pi(8^2) = \pi(580.8) + \pi(64) = \pi(644.8) \approx 2026$ cm² (about 2025 cm² if $l = 26.39$ cm is used).

(d) The frustum is a large cone (slant height L , radius R) with a small cone (slant height $L - l$, radius r) cut off from the top. Curved $SA = \pi RL - \pi r(L - l)$. Because the two cones are similar, $L/R = (L - l)/r$, which gives $rL = R(L - l)$ and so $L = Rl/(R - r)$. Substituting, curved $SA = \pi RL - \pi rL + \pi rl = \pi(R - r)L + \pi rl = \pi Rl + \pi rl = \pi l(R + r)$.

(e) A closed bucket would also need the top circle, so l would add $\pi R^2 = \pi(14^2) = 196\pi \approx 616$ cm² to the total, giving $TSA = \pi(R + r)l + \pi R^2 + \pi r^2$.

Section 3: Pyramids, spheres and hemispheres

(a) A pyramid-shaped gift box has a square base of side $l = 10$ cm and a slant height $s = 13$ cm. Describe the faces on its net, then calculate its total surface area.

(b) In the orange-peel activity, the peel of a sphere was pressed flat onto circles with the same radius as the orange. What did you observe, and what formula for the surface area of a sphere does this give?

(c) Use your answer to (b) to explain how the total surface area of a closed hemisphere, such as the hemispherical tent

(a) The net has one square base and four congruent triangular faces. Base area = $l^2 = 10^2 = 100$ cm². Each triangle has area $\frac{1}{2} \times 10 \times 13 = 65$ cm², so four triangles give $4 \times 65 = 260$ cm². $TSA = l^2 + 2ls = 100 + 2(10)(13) = 100 + 260 = 360$ cm².

(b) The peel covered about four circles that had the same radius as the orange's great circle. So the surface area of a sphere is 4 times the area of its great circle: $SA = 4\pi r^2$.

(c) Cutting the sphere in half gives two equal halves, so the curved surface area of the hemisphere is half of $4\pi r^2$, which is $2\pi r^2$. A closed hemisphere also has a flat circular base of area πr^2 . Total surface area = $2\pi r^2 + \pi r^2 = 3\pi r^2$.

(d) The curved surface area is only the outer curved part. The flat circle is a separate face that exists only when the hemisphere is closed (for example, the floor or base), so it is added only when the total surface area is needed.

at the Maasai Mara camp, is found. (d) Why is the flat circle included in a closed hemisphere's total surface area but not in its curved surface area?	
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Section 4: Settling the shopkeeper's claim

In Lesson 8 you calculated the surface areas of the cylindrical, conical and frustum-shaped buckets, all holding 10 litres (10 000 cm³). (a) The frustum bucket in Section 2 has a volume of about 10 000 cm³. Show how a bucket can hold the same volume as another bucket but still need a different amount of metal. (b) Using the class results from Lesson 8, is the shopkeeper's claim that the frustum bucket uses the most metal correct? Give your reasoning. (c) Is the claim always true for any frustum, cone and cylinder of equal volume? Explain. (d) Frustum-shaped buckets are very common in Kenya. Suggest why manufacturers and customers may still choose them, and what this tells you about how mathematics is used in real life. (e) Name one other solid from your portfolio or from Kenyan life (for example a conical grain store, a frustum water tank or a spherical water tank in Kisumu) and say what surface area calculation would help a builder.	(a) Volume and surface area use different formulae. For example, the frustum with $R = 14$ cm, $r = 8$ cm and $h = 25.7$ cm has volume $\approx 10\,000$ cm³ but needs about 2026 cm² of metal when open. A cone or cylinder with the same 10 000 cm³ has a different radius and height, so its curved surface and base areas come out differently. Changing the shape while keeping the volume fixed changes the surface area. (b) No, the claim is not correct for our 10-litre buckets. When the class calculated all three open buckets using the Bucket Data Sheet, the open cone needed the greatest surface area of sheet metal, more than the frustum. So the evidence does not support the claim that the frustum uses the most metal. (c) No. The surface area of each shape depends on the particular dimensions chosen, such as radius, height and slant height. The result applies to the dimensions used in our comparison and is not a universal rule, so the shopkeeper's claim must be tested by calculation for each set of buckets. (d) A tapered shape lets buckets nest inside each other for stacking and storage, and makes them easy to carry and pour from. Manufacturers balance material cost with strength and practical use, so the cheapest surface area is not the only factor. Mathematics helps people compare options with evidence. (e) For a conical grain store, a builder would calculate the curved surface area πrl (plus the base πr^2 if it is closed) to know how much roofing or sheet metal to buy. For a frustum silo such as the one at Eldoret ($R = 1.5$ m, $r = 0.6$ m, $h = 2.4$ m), $l = \sqrt{(2.4^2 + 0.9^2)} \approx 2.56$ m, and the curved area is $\pi(2.1)(2.56) \approx 16.9$ m².
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Correct use of surface area formulae	Chooses and states the correct formula for every solid (cone, pyramid, sphere, hemisphere, frustum), and correctly distinguishes open from closed shapes and slant height from perpendicular height.	Uses mostly correct formulae with one or two small slips, such as forgetting a base or confusing open and closed.	Uses wrong or missing formulae, or confuses slant height with perpendicular height.
Accurate calculation and units	All calculations are correct and clearly laid out, with correct rounding and units (cm ² , m ²).	Method is correct but there are minor arithmetic or rounding errors, or a unit is missing.	Method is unclear or contains major errors, and units are missing.
Explaining formulae from nets and derivations	Clearly explains where πrl , $\pi(R + r)l$, $4\pi r^2$ and $3\pi r^2$ come from using sectors, annular sectors, cone differences or the orange-peel evidence.	Gives a partly correct explanation of the origin of the formulae, with some steps missing.	States formulae without explaining where they come from.
Using evidence to judge the shopkeeper's claim	Uses the Lesson 8 comparison to conclude that equal volume does not mean equal surface area, explains that the result depends on the dimensions, and does not overgeneralise.	Reaches the correct conclusion but gives limited evidence or does not note that it depends on the dimensions.	Accepts or rejects the claim without evidence from calculation.
Connecting to Kenyan real-life contexts	Links the mathematics to real Kenyan uses (buckets, tanks, silos, tents) and gives sensible reasons why shape choice balances cost, stacking and strength.	Gives a Kenyan example with a simple reason.	Gives no real-life link or an irrelevant one.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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